

MATH 2790 Fall 2025

Homework 3

Due date: Wednesday, Sep 10

Instructions:

- (a) It is expected to show detailed work for every problem. You are allowed to have a discussion with peers on some challenging questions. But then you should work on your own to finish the assignment.
- (b) All questions in this assignment are from sections 1.10–1.11.

Problems:

1. Question (1) on page 76 of textbook under section 1.10. Round to 5 decimal places.
2. The monthly interest rate is 1%, find the equivalent annual effective rate of interest i and the nominal rate of discount convertible quarterly. Round to 5 decimal places.
3. Question (2) on page 77 of textbook under section 1.10.
4. Find the accumulated value of \$2,490 at the end of twelve years if the nominal interest rate was 2% convertible monthly for the first three years, the nominal rate of discount was 3% convertible semiannually for the next two years, the nominal rate of interest was 4.2% convertible quarterly for the next four years, and the annual effective rate of discount was 0.058 for the last three years. Round to 4 decimal places.
5. Question (1) on page 77 of textbook under section 1.11. Round to 5 decimal places.
6. Question (3) on page 77 of textbook under section 1.11.